

PLANET STORIES

20c

STRANGE ADVENTURES ON OTHER
WORLDS—THE UNIVERSE OF
FUTURE CENTURIES

CITADEL OF LOST SHIPS

A Novellet of Space Revens
by LEIGH BRACKETT

SLAVES OF THE NINTH MOON

by
ROSS ROCKLYNNE
NELSON S. BOND
R. R. WINTERBOTHAM
HASSE & DE PINA



PLANET STORIES 20c

STRANGE ADVENTURES ON OTHER
WORLDS—THE UNIVERSE OF
FUTURE CENTURIES

CITADEL OF LOST SHIPS

A Novellet of Space Revers
by LEIGH BRACKETT

SLAVES OF THE NINTH MOON

by
ROSS ROCKLYNNE
NELSON S. BOND
R. E. WINTERBOTHAM
HASSE & DE PINA



The Project Gutenberg eBook of Trouble on Tycho

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Trouble on Tycho

Author: Nelson S. Bond

Illustrator: Walker

Release date: May 28, 2020 [eBook #62260]

Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/62260

Credits: Produced by Greg Weeks, Mary Meehan and the Online
Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK TROUBLE ON
TYCHO ***

TROUBLE ON TYCHO

By NELSON S. BOND

Isobar and his squeeze-pipes were the bane of the Moon Station's existence. But there came the day when his comrades found that the worth of a man lies sometimes in his nuisance value.

[Transcriber's Note: This etext was produced from Planet Stories March 1943.

Extensive research did not uncover any evidence that the U.S. copyright on this publication was renewed.]

The audiophone buzzed thrice—one long, followed by two shorts—and Isobar Jones pressed the stud activating its glowing scanner-disc.

"Hummm?" he said absent-mindedly.

The selenoplate glowed faintly, and the image of the Dome Commander appeared.

"Report ready, Jones?"

"Almost," acknowledged Isobar gloomily. "It prob'ly ain't right, though. How anybody can be expected to get *anything* right on this dagnabbed hunk o' green cheese—"

"Send it up," interrupted Colonel Eagan, "as soon as you can. Sparks is making Terra contact now. That is all."

"That ain't all!" declared Isobar indignantly. "How about my bag—?"

It *was all*, so far as the D.C. was concerned. Isobar was talking to himself. The plate dulled. Isobar said, "Nuts!" and returned to his duties. He jotted neat ditto marks under the word "Clear" which, six months ago, he had placed beneath the column headed: *Cond. of Obs.* He noted the proper figures under the headings *Sun Spots: Max Freq.—Min. Freq.*; then he sketched careful curves in blue and red ink upon the Mercator projection of Earth which was his daily work sheet.

This done, he drew a clean sheet of paper out of his desk drawer, frowned thoughtfully at the tabulated results of his observations, and began writing.

"*Weather forecast for Terra*," he wrote, his pen making scratching sounds.

The audiophone rasped again. Isobar jabbed the stud and answered without looking.

"O.Q.," he said wearily. "O.Q. I told you it would be ready in a couple o' minutes. Keep your pants on!"

"I—er—I beg your pardon, Isobar?" queried a mild voice.

Isobar started. His sallow cheeks achieved a sickly salmon hue. He blinked nervously.

"Oh, jumpin' jimminy!" he gulped. "You, Miss Sally! Golly—'scuse me! I didn't realize—"

The Dome Commander's niece giggled.

"That's all right, Isobar. I just called to ask you about the weather in Oceania Sector 4B next week. I've got a swimming date at Waikiki, but I won't make the shuttle unless the weather's going to be nice."

"It is," promised Isobar. "It'll be swell all weekend, Miss Sally. Fine sunshiny weather. You can go."

"That's wonderful. Thanks so much, Isobar."

"Don't mention it, ma'am," said Isobar, and returned to his work.

South America. Africa. Asia. Pan-Europa. Swiftly he outlined the meteorological prospects for each sector. He enjoyed this part of his job. As he wrote forecasts for each area, in his mind's eye he saw himself enjoying such pastimes as each geographical division's terrain rendered possible.

If home is where the heart is, Horatio Jones—known better as "Isobar" to his associates at the Experimental Dome on Luna—was a long, long way from home. His lean, gangling frame was immured, and had been for six tedious Earth months, beneath the *impervite* hemisphere of Lunar III—that frontier outpost which served as a rocket refueling station, teleradio transmission point and meteorological base.

"Six solid months! Six sad, dreary months!" thought Isobar, "Locked up in an airtight Dome like—like a goldfish in a glass bowl!" Sunlight? Oh, sure! But filtered through ultraviolet wave-traps so it could not burn, it left the skin pale and lustreless and clammy as the belly of a toad. Fresh air? Pooh! Nothing but that everlasting sickening, scented, reoxygenated stuff gushing from atmo-conditioning units.

Excitement? Adventure? The romance he had been led to expect when he signed on for frontier service? Bah! Only a weary, monotonous, routine existence.

"A pain!" declared Isobar Jones. "That's what it is; a pain in the stummick. Not even allowed to—Yeah?"

It was Sparks, audioing from the Dome's transmission turret. He said, "Hyah, Jonesy! How comes with the report?"

"Done," said Isobar. "I was just gettin' the sheets together for you."

"O.Q. But just bring *it*. Nothing else."

Isobar bridled.

"I don't know what you're talkin' about."

"Oh, no? Well, I'm talking about that squawk-filled doodlesack of yours, sonny boy. Don't bring that bag-full of noise up here with you."

Isobar said defiantly, "It ain't a doodlesack. It's a bagpipe. And I guess I can play it if I want to—"

"Not," said Sparks emphatically, "in *my* cubby! I've got sensitive eardrums. Well, stir your stumps! I've got to get the report rolling quick today. Big doings up here."

"Yeah? What?"

"Well, it's Roberts and Brown—"

"What about 'em?"

"They've gone Outside to make foundation repairs."

"Lucky stiffs!" commented Isobar ruefully.

"Lucky, no. Stiffs, maybe—if they should meet any Grannies. Well, scoot along. I'm on the ether in four point sixteen minutes."

"Be right up," promised Isobar, and, sheets in hand, he ambled from his cloistered cell toward the central section of the Dome.

He didn't leave Sparks' turret after the sheets were delivered. Instead, he hung around, fidgeting so obtrusively that Riley finally turned to him in sheer exasperation.

"Sweet snakes of Saturn, Jonesy, what's the trouble? Bugs in your britches?"

Isobar said, "H-huh? Oh, you mean—Oh, thanks, no! I just thought mebbe you wouldn't mind if I—well—er—"

"I get it!" Sparks grinned. "Want to play peekaboo while the contact's open, eh? Well, O.Q. Watch the birdie!"

He twisted dials, adjusted verniers, fingered a host of incomprehensible keys. Current hummed and howled. Then a plate before him cleared, and the voice of the Earth operator came in, enunciating with painstaking clarity:

"Earth answering Luna. Earth answering Luna's call. Can you hear me, Luna? Can you hear—?"

"I can not only hear you," snorted Riley, "I can see you and smell you, as well. Stop hamming it, stupid! You're lousing up the earth!"

The now-visible face of the Earth radioman drew into a grimace of displeasure.

"Oh, it's *you*? Funny man, eh? Funny man Riley?"

"Sure," said Riley agreeably. "I'm a scream. Four-alarm Riley, the cosmic comedian—didn't you know? Flick on your dictacoder, oyster-puss; here's the weather report." He read it. "*Weather forecast for Terra, week of May 15-21—*"

"Ask him," whispered Isobar eagerly. "Sparks, don't forget to ask him!"

Riley motioned for silence, but nodded. He finished the weather report, entered the Dome Commander's log upon the Home Office records, and dictated a short entry from the Luna Biological Commission. Then:

"That is all," he concluded.

"O.Q.," verified the other radioman. Isobar writhed anxiously, prodded Riley's shoulder.

"Ask him, Sparks! Go on ask him!"

"Oh, cut jets, will you?" snapped Sparks. The Terra operator looked startled.

"How's that? I didn't say a word—"

"Don't be a dope," said Sparks, "you dope! I wasn't talking to you. I'm entertaining a visitor, a refugee from a cuckoo clock. Look, do me a favor, chum? Can you twist your mike around so it's pointing out a window?"

"What? Why—why, yes, but—"

"Without buts," said Sparks grumpily. "Yours not to reason why; yours but to do or don't. Will you do it?"

"Well, sure. But I don't understand—" The silver platter which had mirrored the radioman's face clouded as the Earth operator twirled the inonoscope. Walls and desks of an ordinary broadcasting office spun briefly into view; then the plate reflected a glimpse of an Earthly landscape. Soft blue sky warmed by an atmosphere-shielded sun ... green trees firmly rooted in still-greener grass ... flowers ... birds ... people....

"Enough?" asked Sparks.

Isobar Jones awakened from his trance, eyes dulling. Reluctantly he nodded. Riley stared at him strangely, almost gently. To the other radioman, "O.Q., pal," he said. "Cut!"

"Cut!" agreed the other. The plate blanked out.

"Thanks, Sparks," said Isobar.

"Nothing," shrugged Riley "*He twisted* the mike; not me. But—how come you always want to take a squint at Earth when the circuit's open, Jonesy? Homesick?"

"Sort of," admitted Isobar guiltily.

"Well, hell, aren't we all? But we can't leave here for another six months at least. Not till our tricks are up. I should think it'd only make you feel worse to see Earth."

"It ain't Earth I'm homesick for," explained Isobar. "It's—well, it's the things that go with it. I mean things like grass and flowers and trees."

Sparks grinned; a mirthless, lopsided grin.

"We've got *them* right here on Luna. Go look out the tower window, Jonesy. The Dome's nestled smack in the middle of the prettiest, greenest little

valley you ever saw."

"I know," complained Isobar. "And that's what makes it even worse. All that pretty, soft, green stuff Outside—and we ain't allowed to go out in it. Sometimes I get so mad I'd like to—"

"To," interrupted a crisp voice, "what?"

Isobar spun, flushing; his eyes dropped before those of Dome Commander Eagan. He squirmed.

"N-nothing, sir. I was only saying—"

"I heard you, Jones. And please let me hear no more of such talk, sir! It is strictly forbidden for anyone to go Outside except in cases of absolute necessity. Such labor as caused Patrolmen Brown and Roberts to go, for example—"

"Any word from them yet, sir?" asked Sparks eagerly.

"Not yet. But we're expecting them to return at any minute now. Jones! Where are *you* going?"

"Why—why, just back to my quarters, sir."

"That's what I thought. And what did you plan to do there?"

Isobar said stubbornly, "Well, I sort of figured I'd amuse myself for a while —"

"I thought that, too. And with *what*, pray, Jones?"

"With the only dratted thing," said Isobar, suddenly petulant, "that gives me any fun around this dagnabbed place! With my bagpipe."

Commander Eagan said, "You'd better find some new way of amusing yourself, Jones. Have you read General Order 17?"

Isobar said, "I seen it. But if you think—"

"It says," stated Eagan deliberately, "*In order that work or rest periods of the Dome's staff may not be disturbed, it is hereby ordered that the playing*

or practicing of all or any musical instruments must be discontinued immediately. By order of the Dome Commander,' That means you, Jones!"

"But, dingbust it!" keened Isobar, "it don't disturb nobody for me to play my bagpipes! I know these lunks around here don't appreciate good music, so I always go in my office and lock the door after me—"

"But the Dome," pointed out Commander Eagan, "has an air-conditioning system which can't be shut off. The ungodly moans of your—er—so-called musical instrument can be heard through the entire structure."

He suddenly seemed to gain stature.

"No, Jones, this order is final! You cannot disrupt our entire organization for your own—er—amusement."

"But—" said Isobar.

"No!"

Isobar wriggled desperately. Life on Luna was sorry enough already. If now they took from him the last remaining solace he had, the last amusement which lightened his moments of freedom—

"Look, Commander!" he pleaded, "I tell you what I'll do. I won't bother nobody. I'll go Outside and play it—"

"Outside!" Eagan stared at him incredulously. "Are you mad? How about the Grannies?"

Isobar knew all about the Grannies. The only mobile form of life found by space-questing man on Earth's satellite, their name was an abbreviation of the descriptive one applied to them by the first Lunar employers: Granitebacks. This was no exaggeration; if anything, it was an understatement. For the Grannies, though possessed of certain low intelligence, had quickly proven themselves a deadly, unyielding and implacable foe.

Worse yet, they were an enemy almost indestructible! No man had ever yet brought to Earth laboratories the carcass of a Grannie; science was completely baffled in its endeavors to explain the composition of Graniteback physiology—but it was known, from bitter experience, that the carapace or exoskeleton of the Grannies was formed of something harder

than steel, diamond, or battleplate! This flesh could be penetrated by no weapon known to man; neither by steel nor flame, by electronic nor ionic wave, nor by the lethal, newly discovered atomo-needle dispenser.

All this Isobar knew about the Grannies. Yet:

"They ain't been any Grannies seen around the Dome," he said, "for a 'coon's age. Anyhow, if I seen any comin', I could run right back inside—"

"No!" said Commander Eagan flatly. "Absolutely, *no*! I have no time for such nonsense. You know the orders—obey them! And now, gentlemen, good afternoon!"

He left. Sparks turned to Isobar, grinning.

"Well," he said, "one man's fish—hey, Jonesy? Too bad you can't play your doodlesack any more, but frankly, I'm just as glad. Of all the awful screeching wails—"

But Isobar Jones, generally mild and gentle, was now in a perfect fury. His pale eyes blazed, he stomped his foot on the floor, and from his lips poured a stream of such angry invective that Riley looked startled. Words that, to Isobar, were the utter dregs of violent profanity.

"Oh, dagnab it!" fumed Isobar Jones. "Oh, tarnation and dingbust! Oh—*fiddlesticks*!"

II

"And so," chuckled Riley, "he left, bubbling like a kettle on a red-hot oven. But, boy! was he ever mad! Just about ready to bust, he was."

Some minutes had passed since Isobar had left; Riley was talking to Dr. Loesch, head of the Dome's Physics Research Division. The older man nodded commiseratingly.

"It is funny, yes," he agreed, "but at the same time it is not altogether amusing. I feel sorry for him. He is a very unhappy man, our poor Isobar."

"Yeah, I know," said Riley, "but, hell, we all get a little bit homesick now and then. He ought to learn to—"

"Excuse me, my boy," interrupted the aged physicist, his voice gentle, "it is not mere homesickness that troubles our friend. It is something deeper, much more vital and serious. It is what my people call: *weltschmerz*. There is no accurate translation in English. It means 'world sickness,' or better, 'world weariness'—something like that but intensified a thousandfold.

"It is a deeply-rooted mental condition, sometimes a dangerous frame of mind. Under its grip, men do wild things. Hating the world on which they find themselves, they rebel in curious ways. Suicide ... mad acts of valor ... deeds of cunning or knavery...."

"You mean," demanded Sparks anxiously, "Isobar ain't got all his buttons?"

"Not that exactly. He is perfectly sane. But he is in a dark morass of despair. He may try *anything* to retrieve his lost happiness, rid his soul of its dark oppression. His world-sickness is like a crying hunger—By the way, where is he now?"

"Below, I guess. In his quarters."

"Ah, good! Perhaps he is sleeping. Let us hope so. In slumber he will find peace and forgetfulness."

But Dr. Loesch would have been far less sanguine had some power the "giftie gi'en" him of watching Isobar Jones at that moment.

Isobar was not asleep. Far from it. Wide awake and very much astir, he was acting in a singularly sinister role: that of a slinking, furtive culprit.

Returning to his private cubicle after his conversation with Dome Commander Eagan, he had stalked straightway to the cabinet wherein was encased his precious set of bagpipes. These he had taken from their pegs, gazed upon defiantly, and fondled with almost parental affection.

"So I can't play you, huh?" he muttered darkly. "It disturbs the peace o' the dingfounded, dumblasted Dome staff, does it? Well, we'll see about that!"

And tucking the bag under his arm, he had cautiously slipped from the room, down little-used corridors, and now he stood before the huge

impervite gates which were the entrance to the Dome and the doorway to Outside.

On all save those occasions when a spacecraft landed in the cradle adjacent the gateway, these portals were doubly locked and barred. But today they had been unbolted that the two maintenance men might venture out. And since it was quite possible that Brown and Roberts might have to get inside in a hurry, their bolts remained drawn. Sole guardian of the entrance was a very bored Junior Patrolman.

Up to this worthy strode Isobar Jones, confident and assured, exuding an aura of propriety.

"Very well, Wilkins," he said. "I'll take over now. You may go to the meeting."

Wilkins looked at him bewilderedly.

"Huh? Whuzzat, Mr. Jones?"

Isobar's eyebrows arched.

"You mean you haven't been notified?"

"Notified of *what*?"

"Why, the general council of all Patrolmen! Weren't you told that I would take your place here while you reported to G.H.Q.?"

"I ain't," puzzled Wilkins, "heard nothing about it. Maybe I ought to call the office, maybe?"

And he moved the wall-audio. But Isobar said swiftly. "That—er—won't be necessary, Wilkins. My orders were plain enough. Now, you just run along. I'll watch this entrance for you."

"We-e-ell," said Wilkins, "if you say so. Orders is orders. But keep a sharp eye out, Mister Jones, in case Roberts and Brown should come back sudden-like."

"I will," promised Isobar, "don't worry."

Wilkins moved away. Isobar waited until the Patrolman was completely out of sight. Then swiftly he pulled open the massive gate, slipped through, and closed it behind him.

A flood of warmth, exhilarating after the constantly regulated temperature of the Dome, descended upon him. Fresh air, thin, but fragrant with the scent of growing things, made his pulses stir with joyous abandon. He was Outside! He was Outside, in good sunlight, at last! After six long and dreary months!

Raptly, blissfully, all thought of caution tossed to the gentle breezes that ruffled his sparse hair, Isobar Jones stepped forward into the lunar valley....

How long he wandered thus, carefree and utterly content, he could not afterward say. It seemed like minutes; it must have been longer. He only knew that the grass was green beneath his feet, the trees were a lacy network through which warm sunlight filtered benevolently, the chirrupings of small insects and the rustling whisper of the breezes formed a tiny symphony of happiness through which he moved as one charmed.

It did not occur to him that he had wandered too far from the Dome's entrance until, strolling through an enchanting flower-decked glade, he was startled to hear—off to his right—the sharp, explosive bark of a Haemholtz ray pistol.

He whirled, staring about him wildly, and discovered that though his meandering had kept him near the Dome, he had unconsciously followed its hemispherical perimeter to a point nearly two miles from the Gateway. By the placement of ports and windows, Isobar was able to judge his location perfectly; he was opposite that portion of the structure which housed Sparks' radio turret.

And the shooting? That could only be—

He did not have to name its reason, even to himself. For at that moment, there came racing around the curve of the Dome a pair of figures, Patrolmen clad in fatigue drab. Roberts and Brown. Roberts was staggering, one foot dragged awkwardly as he ran; Brown's left arm, bloodstained from shoulder to elbow, hung limply at his side, but in his good right fist he held a spitting Haemholtz with which he tried to cover his comrade's sluggish retreat.

And behind these two, grim, grey, gaunt figures that moved with astonishing speed despite their massive bulk, came three ... six ... a dozen of those lunarites whom all men feared. The Grannies!

III

Simultaneously with his recognition of the pair, Joe Roberts saw him. A gasp of relief escaped the wounded man.

"Jones! Thank the Lord! Then you picked up our cry for help? Quick, man—where is it? There's not a moment to waste!"

"W-where," faltered Isobar feebly, "is *what*?"

"The tank, of course! Didn't you hear our telecast? We can't possibly make it back to the gate without an armored car. My foot's broken, and—" Roberts stopped suddenly, an abrupt horror in his eyes. "You don't have one! You're here *alone*! Then you didn't pick up our call? But, why—?"

"Never mind that," snapped Isobar, "now!" Placid by nature, he could move when urgency drove. His quick mind saw the immediateness of their peril. Unarmed, he could not help the Patrolmen fight a delaying action against their foes, nor could he hasten their retreat. Anyway, weapons were useless, and time was of the essence. There was but one temporary way of staving off disaster. "Over here ... this tree! Quick! Up you go! Give him a lift, Brown—There! That's the stuff!"

He was the last to scramble up the gnarled bole to a tentative leafy sanctuary. He had barely gained the security of the lowermost bough when a thundering crash resounded, the sturdy trunk trembled beneath his clutch. Stony claws gouged yellow parallels in the bark scant inches beneath one kicking foot, then the Granny fell back with a thud. The Graniteback was *not* a climber. It was far too ungainly, much too weighty for that.

Roberts said weakly, "Th-thanks, Jonesy! That was a close call."

"That goes for me, too, Jonesy," added Brown from an upper bough. "But I'm afraid you just delayed matters. This tree's O.Q. as long as it lasts, but—" He stared down upon the gathering knot of Grannies unhappily—"it's

not going to last long with that bunch of superdreadnaughts working out on it! Hold tight, fellows! Here they come!"

For the Grannies, who had huddled for a moment as if in telepathic consultation, now joined forces, turned, and as one body charged headlong toward the tree. The unified force of their attack was like the shattering impact of a battering ram. Bark rasped and gritted beneath the besieged men's hands, dry leaves and twigs pelted about them in a tiny rain, tormented fibrous sinews groaned as the aged forest monarch shuddered in agony.

Desperately they clung to their perches. Though the great tree bent, it did not break. But when it stopped trembling, it was canted drunkenly to one side, and the erstwhile solid earth about its base was broken and cracked—revealing fleshy tentacles uprooted from ancient moorings!

Brown stared at this evidence of the Grannies' power with terror-fascinated eyes. His voice was none too firm.

"Lord! Piledrivers! A couple more like that—"

Isobar nodded. He knew what falling into the clutch of the Grannies meant. He had once seen the grisly aftermath of a Graniteback feast. Even now their adversaries had drawn back for a second attack. A sudden idea struck him. A straw of hope at which he grasped feverishly.

"You telecast a message to the Dome? Help should be on the way by now. If we can just hold out—"

But Roberts shook his head.

"We sent a message, Jonesy, but I don't think it got through. I've just been looking at my portable. It seems to be busted. Happened when they first attacked us, I guess. I tripped and fell on it."

Isobar's last hope flickered out.

"Then I—I guess it won't be long now," he mourned. "If we could have only got a message through, they would have sent out an armored car to pick us

up. But as it is—"

Brown's shrug displayed a bravado he did not feel.

"Well, that's the way it goes. We knew what we were risking when we volunteered to come Outside. This damn moon! It'll never be worth a plugged credit until men find some way to fight those murderous stones-on-legs!"

Roberts said, "That's right. But what are *you* doing out here, Isobar? And why, for Pete's sake, the bagpipes?"

"Oh—the pipes?" Isobar flushed painfully. He had almost forgotten his original reason for adventuring Outside, had quite forgotten his instrument, and was now rather amazed to discover that somehow throughout all the excitement he had held onto it. "Why, I just happened to—Oh! *the pipes!*"

"Hold on!" roared Roberts. His warning came just in time. Once more, the three tree-sitters shook like dried peas in a pod as their leafy refuge trembled before the locomotive onslaught of the lunar beasts. This time the already-exposed roots strained and lifted, several snapped; when the Grannies again withdrew, complacently unaware that the "lethal ray" of Brown's Haemholtz was wasting itself upon their adamant hides in futile fury, the tree was bent at a precarious angle.

Brown sobbed, not with fear but with impotent anger, and in a gesture of enraged desperation, hurled his now-empty weapon at the retreating Grannies.

"No good! Not a damn bit of good! Oh, if there was only some way of fighting those filthy things—"

But Isobar Jones had a one-track mind. "The pipes!" he cried again, excitedly. "That's the answer!" And he drew the instrument into playing position, bag cuddled beneath one arm-pit, drones stiffly erect over his shoulder, blow-pipe at his lips. His cheeks puffed, his breath expelled. The giant lung swelled, the chaunter emitted its distinctive, fearsome, "*Kaa-aa-o-o-o-oro-oong!*"

Roberts moaned.

"Oh, Lord! A guy can't even die in peace!"

And Brown stared at him hopelessly.

"It's no use, Isobar. You trying to scare them off? They have no sense of hearing. That's been proven—"

Isobar took his lips from the reed to explain.

"It's not that. I'm trying to rouse the boys in the Dome. We're right opposite the atmosphere-conditioning-unit. See that grilled duct over there? That's an inhalation-vent. The portable transmitter's out of order, and our voices ain't strong enough to carry into the Dome—but the sound of these pipes is! And Commander Eagan told me just a short while ago that the sound of the pipes carries all over the building!

"If they hear this, they'll get mad because I'm disobeyin' orders. They'll start lookin' for me. If they can't find me inside, maybe they'll look Outside. See that window? That's Sparks' turret. If we can make him look out here—"

"*Stop talking!*" roared Roberts. "Stop talking, guy, and start blowing! I think you've got something there. Anyhow, it's our last hope. *Blow!*"

"And quick!" appended Brown. "For here they come!"



Isobar played, blew with all his might, while the Grannies raged below.

He meant the Grannies. Again they were huddling for attack, once more, a solid phalanx of indestructible, granite flesh, they were smashing down upon the tree.

"*Haa-a-roong!*" blew Isobar Jones.

And—even he could not have foreseen the astounding results of his piping! What happened next was as astonishing as it was incomprehensible. For as the pipes, filled now and primed to burst into whatever substitute for melody they were prodded into, wailed into action—the Grannies' rush came to an abrupt halt!

As one, they stopped cold in their tracks and turned dull, colorless, questioning eyes upward into the tree whence came this weird and vibrant droning!

So stunned with surprise was Isobar that his grip on the pipes relaxed, his lips almost slipped from the reed. But Brown's delighted bellow lifted his paralysis.

"Sacred rings of Saturn-look! They *like* it! Keep playing, Jonesy! Play, boy, like you never played before!"

And Roberts roared, above the skirling of the *piobaireachd* into which Isobar had instinctively swung, "Music hath charms to soothe the savage beast! Then we were wrong. They *can* hear, after all! See that? They're lying down to listen—like so many lambs! Keep playing, Isobar! For once in my life I'm glad to hear that lovely, wonderful music!"

Isobar needed no urging. He, too, had noted how the Grannies' attack had stopped, how every last one of the gaunt grey beasts had suddenly, quietly, almost happily, dropped to its haunches at the base of the tree.

There was no doubt about it; the Grannies *liked* this music. Eyes raptly fixed, unblinking, unwavering, they froze into postures of gentle beatitude. One stirred once, dangerously, as for a moment Isobar paused to catch his breath, but Isobar hastily lipped the blow-pipe with redoubled eagerness, and the Granny relapsed into quietude.

Followed then what, under somewhat different circumstances, should have been a piper's dream. For Isobar had an audience which would not—and in two cases *dared* not—allow him to stop playing. And to this audience he played over and over again his entire repertoire. Marches, flings, dances—the stirring *Rhoderik Dhu* and the lilting *Lassies O'Skye*, the mournful *Coghiegh nha Shie* whose keening is like the sound of a sobbing nation.

The Cock o' the North, he played, and *Mironton ... Wee Flow'r o' Dee* and *MacArthur's March ... La Cucuracha* and—

And his lungs were parched, his lips dry as swabs of cotton. Blood pounded through his temples, throbbing in time to the drone of the chaunter, and a dark mist gathered before his eyes. He tore the blow-pipe from his lips, gasped,

"Keep playing!" came the dim, distant howl of Johnny Brown. "Just a few minutes longer, Jonesy! Relief is on the way. Sparks saw us from his turret window five minutes ago!"

And Isobar played on. How, or what, he did not know. The memory of those next few minutes was never afterward clear in his mind. All he knew was that above the skirling drone of his pipes there came another sound, the metallic clanking of a man-made machine ... an armored tank, sent from the Dome to rescue the beleaguered trio.

He was conscious, then, of a friendly voice shouting words of encouragement, of Joe Roberts calling a warning to those below.

"Careful, boys! Drive the tank right up beneath us so we can hop in and get out of here! Watch the Grannies—they'll be after us the minute Isobar stops playing!"

Then the answer from below. The fantastic answer in Sparks' familiar voice. The answer that caused the bagpipes to slip from Isobar's fingers as Isobar Jones passed out in a dead faint:

"After you? Those Grannies? Hell's howling acres—*those Grannies are stone dead!*"

Afterward, Isobar Jones said weakly, "But—dead? I don't understand. Was it the sound-waves that killed them?"

Commander Eagan said, "No! Grannies absolutely cannot hear. That is one thing we do know about them—though we will soon know a great deal more, now that our biologists have a dozen carcasses to dissect, thanks to you. But Grannies have no auditory apparatus."

"But then—what?" puzzled Isobar. "It couldn't be vibration, because our Patrolmen tried shootin' 'em with the vibro-ray pistol, and nothin' never

happened—"

"Nevertheless," said Dr. Loesch quietly, "it was vibration which killed them, Isobar. That is, of course, only my conjecture, but I believe subsequent study will prove I am correct.

"It was the effect of *dual*, or disharmonic vibration. You see, the vibro-ray pistol expels an ultrasonic wave which disrupts molecular construction sensitive to a single harmonic. The Grannies' composition is more complex. It required the impact of two different wave-lengths, impinging on their nerve centers at the same moment, to destroy them."

"And the bagpipe—" said Isobar with slowly dawning comprehension—"emits two distinct tones at the same time!"

The full meaning of his words flashed upon Isobar. He turned to Commander Eagan, sallow cheeks glowing with new color.

"Then—then what means we've licked our problem!" he cried. "We've found a weapon that'll kill the Grannies, and it won't be necessary to live inside Domes no more! Now we can move out into the open and live like human beings!"

"Absolutely true!" agreed the Commander. "But *you* will not be living Outside, Jones. Not right away, anyway."

"H-uh? W-hat do you mean, Commander?"

"I mean," said Eagan sternly, "that regardless of results, you are still guilty of flagrant disobedience to orders! That, as Commander of this outpost, I cannot tolerate. You are hereby sentenced to thirty days confinement to quarters!"

"But—" stammered Isobar—"but tarnation golly—"

"In the course of which time," continued Commander Eagan imperturbably, "you will serve as Instructor for every man in the Dome—at double salary!"

"You can't *do* me like this!" wailed Isobar. "Jinky-wallopers, I won't—Huh? What's 'at? Instructor? Instructor in *what*?"

"In the—er—art," said Eagan, "of bagpipe playing. If we are to rid Luna of the Grannies, we must all learn how to perform on that—er—lethal weapon.

And, Jones, I think I can truthfully say that this punishment hurts me more than it hurts you!"

*** END OF THE PROJECT GUTENBERG EBOOK TROUBLE ON
TYCHO ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.